

Sebastian Lopez Silva

AI Engineer

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Profile

AI Engineer, with solid experience in software engineering and the **production deployment of AI services** for text, image-based and **multimodal** applications. Particularly interested in **autonomous AI agents**, **LLM systems**, and **production-grade intelligent platforms**.

Education

École des Mines de Paris - PSL, Engineering Degree (M.Sc.) (GPA: 3.7/4.0)	2025 - Expected 2027
Universidad Nacional de Colombia, Computer Engineering (B.Sc.) (GPA: 4.0/4.0)	2021 - Expected 2027

Experience

AI Engineer / Architect Intern - STMicroelectronics (May 2026 – Aug 2026) – Rousset, France (Internship)

- Engineered **RAG** and **Knowledge Graph** ingestion pipelines using vector databases and Google's **Open Knowledge Format (OKF)**, processing >1,000 public documents for semantic retrieval.
- Architected modular **AI agent** context systems and developed the underlying **harness layer** across diverse agent execution frameworks.
- Developed **LLM API services** for AI infrastructure and automated operational workflows using **AI agents** in sandboxed, safe execution environments.
- Conducted systematic **evaluation and benchmarking** of **local LLM infrastructure** to assess latency, throughput, and inference reliability.

AI Engineer / Researcher - Engin AI (Jan 2026 – May 2026) – Miami, United States (Independent Contractor)

- Conducted research, experimentation, and production deployment of an **LLM AI agent**, leveraging deepfake classification models within the agent's **harness layer**.
- Architected custom **long-term memory** and **short-term conversational memory systems** tailored to the project, integrating structured tool execution workflows.
- Designed and executed **internal comparative benchmarks** to evaluate agent decision-making accuracy, inference latency, and detection reliability across multimodal scenarios.

Machine Learning Engineer - Engin AI (Nov 2025 – Jan 2026) – Miami, United States (Independent Contractor)

- Designed and deployed over 5 production-ready **multimodal inference endpoints** (audio, image, video), processing approximately 4,000 daily requests with an average latency of 126 ms.
- Extended **deep learning** architectures for **audio detection**, processing more than 1 million samples and improving experiment reproducibility by 40%.
- Built **scalable data pipelines** for datasets exceeding 100 GB, reducing deployment time by 60% through automated validations and configuration-driven workflows.

Software Engineer - Engin AI (Sept 2025 – Nov 2025) – Miami, United States (Independent Contractor)

- Reduced **ML pipeline** latency by 70-80% by redesigning data preprocessing workflows and optimizing resource utilization in containerized environments.
- Automated annotation validation and debugging workflows (**YOLO/COCO**) via Python scripts, eliminating 75% of manual QA steps and improving dataset reliability.

Junior Software Engineer - Dataconstructors AI (Mar 2025 – Sept 2025) – Bogotá, Colombia

- Reduced geospatial processing time from one week to three hours by designing automated orchestration of Python pipelines interconnected with database-backed services.
- Designed **AI-assisted pipelines** for structured technical documentation and validation workflows.
- Developed Python scripts integrating **LLMs** for **semantic extraction** and automatic validation of critical fields, improving database consistency and decreasing operational errors by 60%.
- Developed and maintained backend services integrating external APIs and relational databases (MySQL/PostgreSQL), applying clean architecture principles to ensure maintainability and scalability.

Skills

Programming Languages: Python • TypeScript • C/C++
Frameworks & Libraries: FastAPI • React • Next.js • Pandas • NumPy • PyTorch • TensorFlow
Databases: SQL (PostgreSQL, SQLite) • NoSQL (Redis, MongoDB) • Vector DBs
Infrastructure & Systems: Linux • Git • GitHub Actions • Testing • CI/CD • Docker • MCP Servers
Cloud: Google Cloud Platform (Compute Engine, Cloud Storage, Vertex AI)
Artificial Intelligence: LLM (OpenAI API, Claude Platform, OpenRouter) • AI Agents & MCP Servers • RAG & Knowledge Graphs (OKF) • Computer Vision • Deep Learning • Local LLM Evaluation
Languages: Spanish (C2) • English (C1) • French (C1) • Portuguese (A2)